LNP™ THERMOCOMP™ COMPOUND TF006

TF-1006
REGION AMERICAS

DESCRIPTION

LNP THERMOCOMP TF006 is a compound based on Polyurethane containing 30% Glass Fiber.

TYPICAL PROPERTY VALUES

Revision 20190816

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, yld, Type I, 5 mm/min	50	MPa	ASTM D 638
Tensile Stress, brk, Type I, 5 mm/min	49	MPa	ASTM D 638
Tensile Strain, yld, Type I, 5 mm/min	16.3	%	ASTM D 638
Tensile Strain, brk, Type I, 5 mm/min	16	%	ASTM D 638
Tensile Modulus, 50 mm/min	1480	MPa	ASTM D 638
Flexural Stress, brk, 1.3 mm/min, 50 mm span	58	MPa	ASTM D 790
Flexural Modulus, 1.3 mm/min, 50 mm span	1690	MPa	ASTM D 790
Tensile Stress, yield, 5 mm/min	51	MPa	ISO 527
Tensile Stress, break, 5 mm/min	51	MPa	ISO 527
Tensile Strain, yield, 5 mm/min	15.5	%	ISO 527
Tensile Strain, break, 5 mm/min	16	%	ISO 527
Tensile Modulus, 1 mm/min	1800	MPa	ISO 527
Flexural Modulus, 2 mm/min	2180	MPa	ISO 178
IMPACT			
Izod Impact, unnotched, 23°C	NB1751	J/m	ASTM D 4812
Izod Impact, notched, 23°C	512	J/m	ASTM D 256
Instrumented Impact Energy @ peak, 23°C	19	J	ASTM D 3763
Multiaxial Impact	17	J	ISO 6603
Izod Impact, unnotched 80*10*4 +23°C	131	kJ/m²	ISO 180/1U
Izod Impact, notched 80*10*4 +23°C	41	kJ/m²	ISO 180/1A
THERMAL			
HDT, 0.45 MPa, 3.2 mm, unannealed	167	°C	ASTM D 648
HDT, 1.82 MPa, 3.2mm, unannealed	78	°C	ASTM D 648
CTE, -40°C to 40°C, flow	2.7E-05	1/°C	ASTM E 831
CTE, -40°C to 40°C, xflow	1.3E-04	1/°C	ASTM E 831
CTE, -40°C to 40°C, flow	2.84E-05	1/°C	ISO 11359-2
CTE, -40°C to 40°C, xflow	1.31E-04	1/°C	ISO 11359-2
HDT/Bf, 0.45 MPa Flatw 80*10*4 sp=64mm	167	°C	ISO 75/Bf
HDT/Af, 1.8 MPa Flatw 80*10*4 sp=64mm	85	°C	ISO 75/Af
PHYSICAL			
Density	1.47	g/cm³	ASTM D 792
Moisture Absorption, 50% RH, 24 hrs	0.36	%	ASTM D 570
Mold Shrinkage, flow, 24 hrs	0.1 – 0.4	%	ASTM D 955

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Mold Shrinkage, xflow, 24 hrs	0.5 – 0.8	%	ASTM D 955
Mold Shrinkage, flow, 24 hrs	0.15	%	ISO 294
Mold Shrinkage, xflow, 24 hrs	0.4	%	ISO 294
Density	1.47	g/cm^3	ISO 1183
Moisture Absorption (23°C / 50% RH)	0.5	%	ISO 62
INJECTION MOLDING			
Drying Temperature	95 – 105	°C	
Drying Time	2	hrs	
Maximum Moisture Content	0.03	%	
Melt Temperature	210	°C	
Nozzle Temperature	205 – 225	°C	
Front - Zone 3 Temperature	200 – 220	°C	
Middle - Zone 2 Temperature	195 – 215	°C	
Rear - Zone 1 Temperature	195 – 210	°C	
Mold Temperature	15 – 45	°C	
Back Pressure	0.2 – 0.3	MPa	
Screw Speed	30 – 60	rpm	
Shot to Cylinder Size	40 – 80	%	

DISCLAIMER